

Specialist in IC Design for Logic, Memory, & Mixed-Signal Systems

Expert in translating system requirements into robust IC designs for logic, memory, and mixed-signal applications.
Adept at leveraging EDA tools for design, verification, and layout.

Education

- 08/2022–
Present **Ph.D. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00 | NeuroAI Lab | Advisor: Dr. Abhronil Sengupta | Expected Graduation: 05/2027
- 08/2022–
08/2024 **M.S. in Electrical Engineering**, *Pennsylvania State University*, University Park, PA, USA, CGPA: 4.00/4.00 | Thesis: Neuromorphic Computing for Lifelong Learning
- 02/2015–
04/2019 **B.Sc. in Electrical and Electronic Engineering**, *Bangladesh University of Engineering & Technology (BUET)*, Dhaka, Bangladesh, CGPA: 3.81/4.00

Industrial Experience

- 09/2026–
Present **NPU & AI Accelerator Research Intern**, *GlobalFoundries*, Malta, NY
- Develop fault-injection and evaluation methods to assess how memory bit errors affect quantized neural networks, using Python, ResNet/CIFAR workloads, quantization, and statistical accuracy/sensitivity analysis.
 - Build reproducible experimentation workflows with automated testing, model configurations, controlled random seeds, and structured result generation to study role-aware protection strategies for reliable, energy-efficient AI accelerators.
 - Explore hardware-oriented Edge AI architectures and computing approaches, including hierarchical sensor-proximal processing, spiking neural networks, hyperdimensional computing, compute-in-memory, and non-volatile memories.
- 05/2026–
07/2026 **Machine Learning Engineer Intern**, *Micron Technology*, Richardson, TX
- Model Non-Volatile Memory (NVM) device characteristics (technology-agnostic) to evaluate speedup and energy savings for memory-centric ML inference.
 - Characterize LLM serving performance (TPOT, TTFT) under Prefill-Decode and Attention-FFN disaggregation schemes (PD, AFD) for hyperscale inference deployment.
 - Investigate quantization error and CIM analog error mitigation techniques to preserve model accuracy under analog compute noise and limited ADC precision.
- 05/2025–
07/2025 **Graduate Technical Intern**, *Intel Corporation*, Hillsboro, OR
- Optimized thin film deposition processes affecting microprocessor circuit performance and electrical characteristics for advanced logic devices.
 - Designed DOE to control critical dimensions (CD) impacting yield and electrical parameters in next-generation processors.
 - Applied statistical modeling to predict process impact on circuit performance metrics and device reliability.

Technical Skills

- Digital/ASIC** RTL design and verification, Verilog/VHDL, Digital circuit simulation (Gate-level, RTL, transistor-level), FPGA prototyping, Synthesis, P&R, DFT, Static timing analysis, Timing closure, CMOS/FinFET circuit design, SRAM/DRAM design and optimization, SRAM circuit fundamentals, sense amplifier design, Systolic Arrays, PIM, memory controller design, VLSI/IC design, ASIC design flow, Digital logic design, Testbench development
- EDA/Sim.** Cadence Virtuoso, Spectre, HSPICE, TCAD, COMSOL, ModelSim (digital simulation), Quartus (FPGA development), Vivado, Synopsys (Design Compiler, PrimeTime, VCS), CIM (Compute-In-Memory) Systems
- Analog Des.** Op-amp and comparator design, Bandgap reference, Biasing circuits, Current mirrors, Differential pairs, Amplifiers, Filters, ADC/DAC, Noise analysis, CIM analog error mitigation
- Programming** Python, CUDA, MATLAB, Verilog, Shell, C/C++, Bash, Linux/Unix
- ML/AI Tools** PyTorch, TensorFlow, TensorRT, Data Visualization (Matplotlib, Seaborn, Plotly), Pandas, NumPy, JMP, Jupyter, LaTeX, AI accelerator development, HPC applications, parallel computing architectures
- AI Dev.** Generative and agentic AI tools (Cursor, Copilot, etc.) for research, design, and code development
- Hardware** Device-circuit co-design, PCB design, Oscilloscope, Signal Generator, Spectrum Analyzer, LabVIEW, Circuit debugging and characterization, NVM device modeling for memory-centric compute
- Collaboration** Git, Microsoft Office, Google Workspace

Academic Research and Teaching Experience

08/2022– **Graduate Research Assistant, NeuroAI Lab, Penn State, University Park, PA**

- Present
- Engineered and validated compact models for emerging Ferroic devices (e.g. FeFETs) in MATLAB and Verilog-A for integration into standard cell and memory design workflows.
 - Analyzed device-level characteristics of spintronic and FeFET devices to inform the design of robust, low-power logic and memory circuits.

08/2024– **Graduate Teaching Assistant, Penn State, University Park, PA**

- 05/2025
- Taught Cadence Virtuoso (schematic/layout), PDK usage, DRC/LVS, and analog/digital design flows; created hands-on lab modules and guided tool/debug workflows.
 - Supervised Capstone projects for 90+ undergraduate students across communications, electronics, and firmware: supported Raspberry Pi/Arduino development, software-defined radio experiments, PCB design, and system integration end-to-end.

Projects

Astromorphic Transformer Lead Student Researcher, 2022–2025. Developed a bioplausible transformer architecture leveraging neuron-astrocyte interactions to emulate self-attention mechanisms. Incorporated Hebbian and presynaptic plasticities with non-linearities and feedback, achieving superior accuracy and faster convergence on sentiment classification (IMDB), image classification (CIFAR-10), and language modeling (WikiText-2) tasks. See publication: [IEEE TCDS 2025].

Ultra Low Cost Electronic Braille Device First Author, 2022–2024. Designed and fabricated a portable, solenoid-based electronic Braille device using 3D printing and Arduino microcontroller. Achieved ultra-low cost (\$20) and lightweight design (338g) with 100 frames/s refresh rate, enabling real-time text-to-braille translation. Implemented PWM control for low power consumption and USB/Bluetooth connectivity. See publication: [iCACCESS 2024].

MIPS Micro-processor Design Lead Student Researcher, 2018–2019. Designed and implemented a 5-stage pipelined MIPS microprocessor in Verilog with instruction/data memory, forwarding, and hazard detection. Verified functionality through comprehensive simulation and synthesized for FPGA deployment.

Select Publications

- [1] **Md Zesun Ahmed Mia**, Malyaban Bal, and Abhronil Sengupta. “RMAAT: Astrocyte-Inspired Memory Compression and Replay for Efficient Long-Context Transformers”. In: *International Conference on Learning Representations (ICLR)*. 2026. URL: <https://openreview.net/forum?id=sTkJdbVxsI>.
- [2] **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Trilinear Compute-in-Memory Architecture for Energy-Efficient Transformer Acceleration”. In: *arXiv preprint arXiv:2604.07628* (2026). URL: <https://arxiv.org/abs/2604.07628>.
- [3] Arnob Saha, **Md Zesun Ahmed Mia**, Jiahui Duan, Kai Ni, and Abhronil Sengupta. “Toward Variation-Tolerant Ferroelectric Neural Computing: Special Session Paper”. In: *2025 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE. 2025, pp. 583–587.
- [4] Tao Zhang, Mingjie Hu, **Md Zesun Ahmed Mia**, Hao Zhang, Wei Mao, Katsuyuki Fukutani, Hiroyuki Matsuzaki, Lingzhi Wen, Cong Wang, Hongbo Zhao, et al. “Self-sensitizable neuromorphic device based on adaptive hydrogen gradient”. In: *Matter 7.5* (2024), pp. 1799–1816.

Recognitions

- Melvin P. Bloom Memorial Outstanding Doctoral Research Award (2026)
- The Wormley Family Graduate Fellowship, Harry G. Miller Fellowships in Engineering (2025)
- Arthur Waynick Graduate Scholarship (2024), Milton and Albertha Langdon Memorial Fellowship (2023), Melvin P. Bloom Memorial Fellowship (2022)

Professional Affiliations

- Reviewer, NeurIPS (2026), IOP Physica Scripta (2026), IEEE MWSCAS (2026)
- Reviewer, IEEE TNNLS (2025), Design Automation Conference (DAC) (2025), IEEE MWSCAS (2025)